

§ 微分方程.

一. 一阶微分方程求解. → 把 y' 解出来

①. $y' = f(x) \cdot g(y)$.

例 15.1

已知 $y = f(x) \Rightarrow \Delta y = \frac{y \Delta x}{1+x^2} + \alpha$, α 比 Δx 高阶的无穷小, $y(0) = \pi$, $y(1) = ?$

① 用定义 $\Delta y = \Delta x + \alpha$
 $\rightarrow \frac{dy}{dx} = \frac{y}{1+x^2}$

② 直接求 $\frac{\Delta y}{\Delta x} = \frac{y}{1+x^2} + \alpha$

$\rightarrow \frac{dy}{y} = \frac{dx}{1+x^2} \rightarrow \ln|y| = \arctan x + \ln C_1$

$|y| = \pm C_1 e^{\arctan x}$

$y = C e^{\arctan x}$

$y(0) = \pi. \pi = C e^0 = C \rightarrow C = \pi$
 $y = \pi e^{\arctan x}$
 $y(1) = \pi e^{\frac{\pi}{4}}$

例 15.2

求 $y' = 1-x+y^2-xy^2$ 通解

$y' = 1-x+(1-x)y^2 = (1-x)(1+y^2)$

$\int \frac{dy}{1+y^2} = \int (1-x) dx$

$\arctan y = x - \frac{x^2}{2} + C$

② ~~齐次型~~ $y' = f(ax+by+c)$

令 $u = ax+by+c \quad u'_x = a + by'_x = a + bf(u)$

例.

求 $dy = \sin(x+y+100) \cdot dx$

令 $x+y+100 = u \quad 1 + \sin(x+y+100) = u'$
 $\rightarrow \frac{du}{1+\sin u} = dx \rightarrow x + C = \int \frac{1-\sin u}{1-\sin^2 u} du$

$x + C = \int \sec^2 u du - \int \sec u \tan u du$

$= \tan u - \sec u$

$\rightarrow \tan(x+y+100) - \sec(x+y+100) = x + C$

③ 齐次型.

1) $y' = f(\frac{y}{x}) \rightarrow \frac{y}{x} = u \rightarrow y = ux$

$\frac{du}{f(u)-u} = \frac{dx}{x} \leftarrow f(u) = y' = u'x + u$

2) $\frac{1}{y'} = f(\frac{x}{y}) \rightarrow \frac{x}{y} = u \rightarrow x = uy$

$\frac{du}{f(u)-u} = \frac{dy}{y} \leftarrow f(u) = \frac{dx}{dy} = \frac{dy}{dy} \cdot y + u \cdot 1$

例 15.3

求微分方程 $(x \frac{dy}{dx} - y) \arctan \frac{y}{x} = x$ 通解.

$$\rightarrow \left(\frac{dy}{dx} - \frac{y}{x} \right) \arctan \frac{y}{x} = 1$$

$$\text{令 } \frac{y}{x} = u \rightarrow \frac{dy}{dx} = \frac{du}{dx} \cdot x + u$$

$$\rightarrow \left[\left(\frac{du}{dx} \cdot x + u \right) - u \right] \arctan u = 1$$

$$\frac{du}{dx} \cdot x = \frac{1}{\arctan u}$$

$$\int \arctan u \cdot du = \int \frac{1}{x} dx$$

$$u \arctan u - \int u \arctan u = \ln|x| + \ln C_1$$

$$u \arctan u - \int \frac{d(u \arctan u)}{1+u^2} = \ln|x| + \ln C_1 = u \arctan u - \frac{1}{2} \ln(1+u^2)$$

$$\rightarrow u \arctan u = \ln C_1 |x| \sqrt{1+u^2}$$

$$\rightarrow \frac{y}{x} \arctan \frac{y}{x} = \ln \left(C \sqrt{1 + \frac{y^2}{x^2}} \right)$$

$$\rightarrow C \sqrt{x^2 + y^2} = e^{\frac{y}{x} \arctan \frac{y}{x}}, C > 0.$$

③ 一阶线性型

$$① y' + p(x)y = q(x).$$

★ 求导公式逆用 $(uv)' \leftarrow u'v + uv'$

例 15.4

过 $(\frac{1}{2}, 0)$ 且满足关系式 $y' \arcsin x + \frac{y}{\sqrt{1-x^2}} = 1$.

$$\left(y' + \frac{y}{\arcsin x \sqrt{1-x^2}} = \frac{1}{\arcsin x} \right) X.$$

$$y' \arcsin x + y \cdot \frac{1}{\sqrt{1-x^2}} = (y \arcsin x)' = 1$$

$$\rightarrow \int y \arcsin x \cdot dx = \int 1 dx \quad y \arcsin x = x + C$$

$$(e^0)' = e^0 \cdot 0'$$

$$(e^0 y)' = e^0 [y' + 0' y] \quad 0' = p(x)$$

$$y = e^{-\int p dx} \left(\int e^{\int p dx} \cdot q dx + C \right) \quad 0 = \int p(x) dx.$$

→ 如果 $\int p dx = \ln$, 不用加绝对值, 其他情况要加绝对值

④ 伯努利方程

$$y' + p(x)y = q(x)$$

$$\rightarrow y' + p(x)y = q(x) \cdot \boxed{y^{-n}} \quad (n \neq 0, n \neq 1)$$

$$\Rightarrow \text{step 1. } y^n y' + p(x) y^{1-n} = q(x).$$

$$\text{令 } y^{1-n} = z$$

$$\text{step 2} \quad z' = (y^{1-n})' = (1-n)y^{-n} \cdot \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow y^{-n} y' + p(x) y^{1-n} = q(x)$$

$$\text{step 3.} \quad \frac{1}{1-n} \frac{dz}{dx} + p(x) \cdot z = q(x)$$

例 15.8.

求微分方程 $y' \cos y = (1 + \cos x \sin y) \sin y$ 通解.

$$\star (\sin y)'_x = (1 + \cos x \sin y) \sin y$$

$$\text{令 } \sin y = z, \quad z' = (1 + z \cos x) z = z + z^2 \cos x$$

$$\left(\frac{z' - z}{y' - q(x)y} = p(x) \cdot (y^n) \right)$$

$$\text{step 1.} \quad z^{-2} \cdot z' - z^{-1} = \cos x$$

$$\text{step 2 令 } z^{-1} = u, \quad u' = -1 z^{-2} \cdot z'$$

$$\rightarrow -u' - u = \cos x$$

$$\text{step 3.} \quad \frac{u' + u}{y' + p y} = q \quad y = e^{-\int p dx} \left(\int e^{\int p dx} q dx + C \right)$$

$$u = e^{-x} \left(-\int e^x \cos x dx + C \right)$$

$$= e^{-x} \left(-e^x \sin x - e^x \cos x + \int e^x \right)$$

$$\left(\begin{array}{c|c|c} e^x & e^x & e^x \\ \cos x & \sin x & -\cos x \end{array} \right) \int e^x \cos x dx = e^x \sin x + e^x \cos x$$

$$\int e^x \cos x dx = \frac{e^x}{2} (\sin x + \cos x)$$

$$\text{原} = e^{-x} \left[-\frac{e^x}{2} (\sin x + \cos x) + C \right]$$

$$= -\frac{\sin x + \cos x}{2} + C e^{-x} = \text{①} = \frac{1}{\sin y}$$

回代.

二. 二阶可降阶微分方程

① 能写成 $y'' = f(x, y')$ 缺 y ← “超杀绝”

$$y' \rightarrow p$$

$$y'' \rightarrow p'$$

$$\text{原方程变为 } \frac{dp}{dx} = f(x, p).$$

例 15.9

$$y'' [x + (y')^2] = y'$$

$$\text{令 } y'' = p', \quad y' = p.$$

$$\text{则 } p'(x + p^2) = p \quad \frac{dp}{dx} = \frac{p}{x + p^2}$$

$$x dp + p^2 dp = p dx$$

$$\frac{p dx - x dp}{p^2} = dp$$

$$\Rightarrow d\left(\frac{x}{p}\right) = dp \Rightarrow \frac{x}{p} = p + C_1 \quad \frac{dy}{dx} = \sqrt{x}$$

$$y(1) = y'(1) = 1. \text{ 附特解}$$

$$y = \frac{2}{3} x^{\frac{3}{2}} + \frac{1}{3}$$

$$C_1 = 0,$$

$$p^2 = x.$$

$$p = \sqrt{x}$$

$$\frac{dy}{dx} = \sqrt{x}$$

$$u = \frac{2}{3} x^{\frac{3}{2}} + C_2$$

② 能写成 $y'' = f(y, y')$ (缺 x 型)
 $\downarrow$
 绝不允许 x 再出现.
 "确实除根".

例 15.10

求 $y'' = \frac{1+(y')^2}{2y}$ 通解.

令 $y' = p$ $y'' = p' = \frac{dp}{dy} \cdot p$

$\rightarrow p \cdot \frac{dp}{dy} = \frac{1+p^2}{2y}$

$\rightarrow \frac{p}{1+p^2} dp = \frac{dy}{2y}$

$\ln(1+p^2) = \ln|y| + \ln C_1$

$1+p^2 = C_1|y| = \pm C_1 y = C_1 y$

$\neq p = y' = \pm \sqrt{C_1 y - 1}$

$\int \frac{dy}{\sqrt{C_1 y - 1}} = \pm \int dx$

$\frac{2\sqrt{C_1 y - 1}}{C_1} = \pm x + C_2$

$2\sqrt{C_1 y - 1} = \pm C_1 x + C_1 C_2 \quad C_1^2(x+C_2)^2$

$4(C_1 y - 1) = \pm C_1^2 x^2 + C_1^2 C_2^2 \pm 2C_1^2 C_2 x$

三. 高阶常系数线性微分方程

① $y'' + py' + qy = f(x)$

step 1. $y'' + py' + qy = 0$

即 $\lambda^2 + p\lambda + q = 0 \rightarrow \lambda_1 \lambda_2$

$\Delta = p^2 - 4q \begin{cases} > 0 \Rightarrow y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} \\ = 0 \Rightarrow y = (C_1 + C_2 x) e^{\lambda x} \\ < 0 \Rightarrow y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] \end{cases}$

step 2. 特解 y^* 代入方程 求待定系数

例 15.11.

$y'' - 2y' + y = 4xe^x$

① $\lambda^2 - 2\lambda + 1 = 0 \quad (\lambda - 1)^2 = 0 \quad \lambda = 1$

$y_h = (C_1 + C_2 x) e^x$

② $y'' - 2y' + y = 4xe^x$

设 $y^* = e^x (ax + b) \cdot x^2$

$\lambda_1 = \lambda_2 = 1 = \alpha$

把 y^* 代入方程

求得 $a = \frac{2}{3}, b = 0$

$y^* = \frac{2}{3} x^2 e^x$

③ $y_{\text{通}} = (C_1 + C_2 x) e^x + \frac{2}{3} x^2 e^x$

$y'' + py' + qy = e^{\alpha x} \cdot P_n(x)$

设 $y^* = e^{\alpha x} \cdot Q_n(x) \cdot x^k$

1) $e^{\alpha x}$ 照抄 2) $Q_n(x)$ 是 x 的 n 次多项式

3) $k = 0, \alpha \neq \lambda_1, \alpha \neq \lambda_2$
 1' $\alpha = \lambda_1$ 或 $\alpha = \lambda_2$
 2' $\alpha = \lambda_1 = \lambda_2$

例 15.12

$$y'' + 4y' + 5y = e^{-2x} \sin x$$

$$\textcircled{1} \lambda^2 + 4\lambda + 5 = 0$$

$$\lambda_{1,2} = \frac{-4 \pm \sqrt{16 - 20}}{2} = -2 \pm i$$

$$y_h = e^{-2x} [C_1 \cos x + C_2 \sin x]$$

$$\textcircled{2} y'' + 4y' + 5y = e^{-2x} \sin x$$

$$= e^{-2x} [0 \cdot \cos x + 1 \cdot \sin x]$$

$$\text{设 } y^* = e^{\alpha x} [A \cos x + B \sin x]$$

$$\therefore y^* = -\frac{1}{2} e^{-2x} \cos x \cdot x$$

$$y'' + py' + qy = e^{\alpha x} [P_m(x) \cos \beta x + P_n(x) \sin \beta x]$$

$$\text{设 } y^* = e^{\alpha x} [\alpha_1^{(1)} \cos \beta x + \alpha_2^{(2)} \sin \beta x] \cdot x^k$$

$$(k = \max\{m, n\})$$

$$k = \begin{cases} 0, & \alpha \pm \beta i \neq \lambda_{1,2} \\ 1, & \alpha \pm \beta i = \lambda_{1,2} \end{cases}$$

$$y'' - y = e^x + 4 \cos x$$

$$\text{step 1. } y'' - y = 0. \quad \Delta = 0 + 4 > 0.$$

$$\lambda^2 - 0\lambda - 1 = \lambda^2 - 1 = 0 \quad \lambda_1 = 1 \quad \lambda_2 = -1.$$

$$\text{step 2. } y'' - y = e^x \cdot x^0.$$

$$y'' - y = 4 \cos x.$$

$$y^* = e^x \cdot A \cdot x$$

$$y^* = e^{0x} [4 \cos x + 0 \sin x]$$

$$[Axe^x]'' - Axe^x = e^x \quad y^* = e^{0x} [B \cos x + C \sin x] \cdot x^0$$

$$\rightarrow A = \frac{1}{2}$$

$$y^* = B \cos x + C \sin x.$$

$$\rightarrow B = -2 \quad C = 0$$

$$\text{step 3. } y = [C_1 e^x + C_2 e^{-x}] + \frac{1}{2} e^x \cdot x + (-2 \cos x)$$

② 能写成 $y'' + py' + qy = f_1(x) + f_2(x)$.

求特解时拆成两个.

$$y'' + py' + qy = f_1(x) \rightarrow y_1^*$$

$$y'' + py' + qy = f_2(x) \rightarrow y_2^*$$

$$\text{通} = y + y_1^* + y_2^*$$

例 15.13

求方程 $y'' - y = e^x + 4 \cos x$ 一条解 (曲线).

s.t. 曲线在 (0, 1) 处与 $y = x - x + 1$ 相切.

② 能写成 $x^2 y'' + pxy' + qy = f(x)$ 欧拉方程

$$\text{令 } x = e^t, \quad t = \ln x \quad \frac{dt}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{1}{x} \cdot \frac{dy}{dt}$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \cdot \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x^2} \frac{d^2 y}{dt^2}$$

$$= \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \frac{1}{x^2}$$

$$\Rightarrow \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) + p \left(\frac{dy}{dt} \right) + qy = f(x)$$

$$\rightarrow \frac{d^2y}{dt^2} + (p-1)\frac{dy}{dt} + qy = f(x)$$

有 $\boxed{x^2 y'' + pxy' + qy = f(x)}$ *

例 15.14.

$$x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = 0. \text{ 求解}$$

$$x=e^t \rightarrow \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0.$$

$$\lambda^2 + 3\lambda + 2 = 0 \quad \lambda_1 = -1 \quad \lambda_2 = -2$$

$$y = C_1 e^{-t} + C_2 e^{-2t}$$

$$\text{换元} = \frac{C_1}{x} + \frac{C_2}{x^2}$$

④ $y^{(n)}$ 情形.

$$\text{如 } y''' + p_1 y'' + p_2 y' + p_3 y = 0.$$

$$\text{即 } \lambda^3 + p_1 \lambda^2 + p_2 \lambda + p_3 = 0 \Rightarrow \lambda_{1,2,3}.$$

例 15.15

求一个以 $y_1 = te^t$, $y_2 = \sin 2t$ 为两个特解的四阶常系数四阶.

$$te^t \rightarrow (C_1 + C_2 t)e^t \rightarrow \text{两重根} \rightarrow \lambda_1 = \lambda_2 = 1$$

$$\sin 2t \rightarrow e^{0x}(C_3 \cos 2t + C_4 \sin 2t) \rightarrow \lambda_{3,4} = 0 \pm 2i$$

$$\Rightarrow (\lambda-1)^2 (\lambda^2 + 4) = 0.$$

$$\Rightarrow \lambda^4 - 2\lambda^3 + 5\lambda^2 - 8\lambda + 4 = 0$$

$$y^{(4)} - 2y''' + 5y'' - 8y' + 4y = 0.$$

例 15.16.

用变量代换 $x = \cos t$ ($0 < t < \pi$) 化简为方程

$$(1-x^2)y'' - xy' + y = 0.$$

并求其满足 $y|_{x=0} = 1$, $y'|_{x=0} = 2$ 特解.

$$\sin^2 t.$$

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = -\frac{1}{\sin t} \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(-\frac{1}{\sin t} \frac{dy}{dt} \right) \frac{1}{\frac{dx}{dt}} = \frac{+1}{\sin^2 t} \left(\frac{\cos t}{\sin^2 t} \frac{dy}{dt} + \frac{1}{\sin t} \frac{d^2y}{dt^2} \right)$$

代入原方程.

$$(1 - \cos t) \left(\frac{1}{\sin^2 t} \frac{d^2 y}{dt^2} - \frac{\cos t}{\sin^3 t} \frac{dy}{dt} \right) + \frac{\cos t}{\sin t} \frac{dy}{dt} + y = 0$$

$$\frac{d^2 y}{dt^2} + y = 0.$$

$$\lambda^2 + 1 = 0 \rightarrow \lambda = \pm i$$

$$\rightarrow y_{\text{通}} = C_1 \cos t + C_2 \sin t$$

$$\text{代入 } y_{\text{通}} = C_1 x + C_2 \sqrt{1-x^2}$$

$$y|_{x=0} = 1 \quad y'|_{x=0} = 2 \rightarrow y = 2x + \sqrt{1-x^2}$$

3. 用因变量换元.

例 15.17. 利用 $y = \frac{u}{\cos x}$ 化简方程

$$y'' \cos x - 2y' \sin x + 3y \cos x = e^x \quad \text{化简并求通解.}$$

$$\begin{aligned} y'_x &\rightarrow u'_x + u' \sec x \tan x \\ y'_x &= u' \sec x + u \sec x \tan x \\ y''_x &= u'' \sec x + 2u' \sec x \tan x + u (\sec x \tan^2 x + \frac{\sec^3 x}{1-x^2}) \end{aligned}$$

$$\text{代入. } u'' + 4u = e^x$$

4 用 x, y 地位互换.

• 例 15.18.

反函数. $dx/dy = 1/(dy/dx)$

$$\frac{d^2 x}{dx^2} + (y + \sin x) \left(\frac{dx}{dy} \right)^2 = 0$$

$$\xrightarrow{dy/dx} \frac{d^2 x}{dy^2} = \frac{d \left(\frac{dx}{dy} \right)}{dy} = \frac{d \left(\frac{dx}{dy} \right)}{dx} \frac{dx}{dy} = - \frac{y''_{xx}}{(y'_x)^3}$$

$$\rightarrow - \frac{y''_{xx}}{(y'_x)^3} + (y + \sin x) \frac{1}{(y'_x)^3} = 0$$

$$-y''_{xx} + y + \sin x = 0$$

$$y'' - y = \sin x$$

五. 应用题.

1. 用极限, 导数定义或积分等式建方程.

例 11.19 用导数定义
 $\lim_{h \rightarrow 0} \left[\frac{f(x+h)}{f(x)} \right]^{\frac{1}{h}} = e^{\frac{1}{x}}, f(x) > 0, \lim_{x \rightarrow +\infty} f(x) = 1.$

$$\text{求 } f(x). \quad \lim_{h \rightarrow 0} \frac{1}{h} \frac{f(x+h) - f(x)}{f(x)} = e^{\frac{1}{x}}$$

$$\begin{aligned} \text{导数定义} \rightarrow &= e^{\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{hx} \cdot \frac{x}{f(x)}} \\ &= e^{\lim_{h \rightarrow 0} \frac{x}{f(x)} \cdot \frac{1}{h} \frac{f(x+h) - f(x)}{h}} = e^{\frac{x}{f(x)} f'(x)} \end{aligned}$$

$$\rightarrow \frac{x}{f(x)} f'(x) = \frac{1}{x} \quad \frac{x}{y} \frac{dy}{dx} = \frac{1}{x}$$

$$\frac{dy}{y} = \frac{dx}{x^2} \quad \ln|y| = -\frac{1}{x} + \ln C$$

$$y = ce^{-\frac{1}{x}}$$

例 11.20 用极限

设 $f(x)$ 在 $(-1, +\infty)$ 上具有连续一阶导数, 且 $f(0) = 1$.

$$\text{及 } f'(x) + f(x) - \frac{1}{x+1} \int_0^x f(t) dt = 0$$

求 $f(x)$, 并证明. 当 $x > 0$ 时, 有 $e^{-x} < f(x) < 1$.

$$(x+1)(f'(x) + f(x)) = \int_0^x f(t) dt$$

两边求导.

$$f'(x) + f(x) + (x+1)[f''(x) + f'(x)] = f(x)$$

$$(x+1)f''(x) + (x+2)f'(x) = 0.$$

$$\leq f'(x) = p \quad f''(x) = p'$$

$$(x+1)p' + (x+2)p = 0. \quad x \in (-1, +\infty) \therefore x+1 \neq 0.$$

$$p' + \frac{x+2}{x+1} p = 0 \Rightarrow p' + (1 + \frac{1}{x+1})p = 0.$$

$$\frac{dp}{dx} = -\frac{x+2}{x+1} p \rightarrow \int \frac{dp}{p} = -\int \frac{x+2}{x+1} dx$$

$$\Rightarrow p = y'(x) = \frac{ce^x}{x+1} > 0. \quad (x > -1).$$

$$\therefore \frac{e^x}{x+1} f(0) = 1 \Rightarrow C = -1$$

$$\therefore y'(x) = \frac{-e^{-x}}{x+1}, \quad x > 0 \text{ 时 } [0, x] \text{ 上 } \frac{e^{-t}}{1+t} < 0.$$

$$\text{为证 } e^{-x} < f(x), \text{ 构造 } F(x) = f(x) - e^{-x}. \quad f(x) - f(0) = x f'(\xi) = -x \frac{e^{-\xi}}{1+\xi} < 0. \quad f(x) < f(0) = 1 \quad (x > 0)$$

$$F'(x) = f'(x) + e^{-x} = -\frac{e^{-x}}{1+x} + e^{-x} = \frac{xe^{-x}}{1+x} > 0, \quad (x > 0)$$

$$\therefore F(x) \uparrow \quad F(x) > F(0) = 0. \quad f(x) > e^{-x}.$$

2. 用几何应用建方程

1) 曲线斜率 $k = f'(x_0) = \tan \alpha$.

2) 公切线 $f'(x_0) = g'(x_0)$.

3) 截距 (两正可负).

$$Y - y = y'(X - x) \begin{cases} \text{令 } Y=0 & X_{\text{截}} = 0 \\ \text{令 } X=0 & Y_{\text{截}} = 0 \end{cases}$$

4) 面积 $\int_a^b f(x) dx$

5) 体积 $V_x = \int_a^b \pi f(x)^2 dx$ $V_y = \int_a^b \pi x^2 f(x) dx$

6) 平均值

7) 弧长

8) 侧面积 $S = \int_a^b 2\pi |y(x)| \sqrt{1+(y')^2} dx$

9) 曲率 $k = \frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}}$

10) 形心.

3. 用变化率建方程.

元素衰变. $\frac{dN}{dt} = -\lambda N$

$$\int \frac{dN}{dt} = -\lambda \int dt$$

人口增长问题 $\int \frac{dN}{dt} = \lambda \int dt$

$$(\text{改进 } \int \frac{dN}{dt} = (r - bN(t)) \int dt)$$

抛物线问题 (追踪问题).

例 15.26

在 xOy 平面上, $|PQ|=1$, 初始时刻 P 在原点, Q 在 $(1,0)$ 点, 若 P 点沿着 y 轴正向移动, 求轨迹?

$P(0, Y)$

$Q(x, y)$

$$\tan \alpha = \frac{Y-y}{0-x}$$

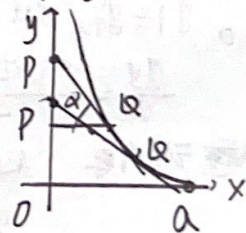
$$\frac{dy}{dx} = \frac{y-Y}{x} = \frac{-\sqrt{1-x^2}}{x}$$

$$y = - \int \frac{\sqrt{1-x^2}}{x} dx$$

$$\text{令 } x = \cos t \quad y = - \int \frac{\sin t}{\cos t} d(\cos t)$$

$$= + \int \frac{1-\cos^2 t}{\cos t} dt = \ln |\sec t + \tan t| - \sin t$$

$$= \ln \left| \frac{1+\sqrt{1-x^2}}{x} \right| - \sqrt{1-x^2}$$



例15.27

设位于坐标轴原点的甲追踪位于 x 轴上点 $A(1,0)$ 处的乙, 甲始终对准乙, 乙以匀速 v_0 沿 y 轴前进, 甲速 $= kv_0$, $k > 0$, 设甲追踪乙曲线方程 $y = y(x)$.

(1) 证明 $y = y(x)$ 满足 $k(1-x)y'' = \sqrt{1+(y')^2}$

且 $y(0) = 0, y'(0) = 0$.

(2) k 为何值时, 甲可追上乙, 并求出甲追上乙的时刻.

[solution] $v_{\text{甲}} = kv_0 \Rightarrow S_{\text{甲}} = kS_{\text{乙}}$

$AP(x, y)$.

$$\begin{cases} \int_0^x \sqrt{1+(y')^2} dx = kv_0 t \\ \frac{dy}{dx} = \frac{v_0 t - y}{1-x} \end{cases}$$

整理得 $k(1-x)y'' = \sqrt{1+(y')^2}$,

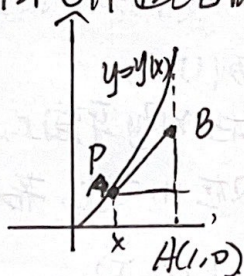
$y(0) = 0, y'(0) = 0$.

(2) 令 $P(x) = y'(x), P'(x) = y''(x)$

$$\rightarrow k(1-x)P' = \sqrt{1+P^2} \quad k(1-x) \frac{dP}{dx} = \sqrt{1+P^2}$$

$$\int \frac{k(1-x)P'}{\sqrt{1+P^2}} dx = \int \frac{dP}{k(1-x)}$$

$$\int \frac{dP}{\sqrt{1+P^2}} = \int \frac{dx}{k(1-x)}$$



$$\ln(P + \sqrt{1+P^2}) = -\frac{1}{k} \ln(1-x) + \ln C$$

$$\rightarrow P + \sqrt{1+P^2} = C(1-x)^{-\frac{1}{k}}$$

$$\rightarrow P + \sqrt{1+P^2} = (1-x)^{-\frac{1}{k}} \quad (1)$$

化简方法 $\frac{1}{P + \sqrt{1+P^2}} = (1-x)^{\frac{1}{k}}$

$$\frac{P - \sqrt{1+P^2}}{P^2 - (1+P^2)} = \frac{1}{\sqrt{1+P^2}} - P = (1-x)^{\frac{1}{k}} \quad (2)$$

$$\Rightarrow \frac{(1) - (2)}{2} \quad P = \frac{(1-x)^{-\frac{1}{k}} - (1-x)^{\frac{1}{k}}}{2}$$

$$\rightarrow 2y' = (1-x)^{-\frac{1}{k}} - (1-x)^{\frac{1}{k}}$$

$$(1) \quad k=1 \text{ 时 } y(x) = -\frac{1}{2} \ln(1-x) + \frac{(1-x)^2}{4} - \frac{1}{4}$$

$\lim_{x \rightarrow 1} y(x) = +\infty$. 甲追不上乙

$$(2) \quad k \neq 1 \text{ 时 } y(x) = \frac{k}{2} \left[\frac{1}{k+1} (1-x)^{\frac{k+1}{k}} - \frac{1}{k-1} (1-x)^{\frac{k-1}{k}} \right] + \frac{k}{k^2-1}$$

若 $k < 1$, $\lim_{x \rightarrow 1} y(x) = +\infty$

若 $k > 1$, $\lim_{x \rightarrow 1} y(x) = \frac{k}{k^2-1}$

当 $k > 1$ 时, 甲在 $(1, \frac{k}{k^2-1})$ 追上乙.